Performance Testing

Functional, latency and resilience test strategy

Project: AI WeatherWise API   Track: AI-ML & GEN-AI Track

Prepared: 07 October 2026   Team ID: Not provided

Maximum Marks: 5

Team Members & Assigned Roles

Team Member	Role
Jai Bharath A	Team Lead & Lead Backend/API Architect
Mukesh Balaji	Generative AI & Prompt Engineering Engineer
Yadagiri Mani	Database & Data Engineering Specialist
Deepak G	Security, Testing & Reliability Engineer
Naveen	API Integration, Documentation & Demonstration Engineer

Test ID	Scenario	Method	Expected result	Owner
PT-01	Register + hash	POST /register	2xx response; password never returned	Deepak
PT-02	Login token	POST /login	JWT issued for valid credentials	Deepak
PT-03	Weather live	GET /weather/:city	Weather JSON returned	Jai
PT-04	Weather outage	Simulate upstream failure	Deterministic mock response	Jai + Deepak
PT-05	AI summary	POST /weather-summary	Natural-language response	Mukesh
PT-06	AI quota/network failure	Block AI dependency	Rule-based summary/advice	Mukesh + Deepak
PT-07	Location isolation	Use user A token on user B record	Access denied / not found	Yadagiri + Deepak
PT-08	Duplicate location	Create same city/country twice	Unique constraint handled	Yadagiri
PT-09	Regression	Run complete Thunder Client collection	All defined requests pass	Naveen

Performance Measures

- Capture average and p95 latency for authentication, weather and AI endpoints.
- Separate upstream latency from local processing latency.
- Repeat AI calls carefully to avoid distorting results through free-tier quota limits.
- Record fallback activation separately; fallback latency is not equivalent to live API latency.

Reliability Test Matrix

Failure	Expected behavior	Severity	Mitigation
OpenWeatherMap unavailable	Mock weather returned	High	Deterministic local generator
Gemini unavailable	Rule-based summary/recommendation	High	Local heuristic engine
MongoDB unavailable	Controlled 5xx response; no silent corruption	High	Centralized error handling + retry strategy in future
Invalid JWT	401/403 response	High	JWT middleware
Duplicate favorite	Validation/duplicate error	Medium	Compound unique index

Test Evidence Principle

Performance claims should be reported from repeatable local test runs and clearly labeled with environment, request count and network conditions. The supplied project documents describe benchmark-style testing; this deliverable keeps the methodology explicit so evaluators can reproduce it.